

Homework 3

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1 Part 0: Grating Coupling Context

For this work, a standard silicon photonic grating coupler was used, as provided in the generic PDK of the design environment. This component enables efficient coupling of light between an optical fiber and on-chip waveguides.

Grating couplers are mandatory for chip characterization in a laboratory setting because they provide a practical interface between optical fibers and submicron waveguides, allowing light to be coupled in and out of the chip without requiring edge coupling or complex alignment procedures. This enables wafer-scale testing and simplifies experimental measurements [2]. (See figure 1 for an example.)

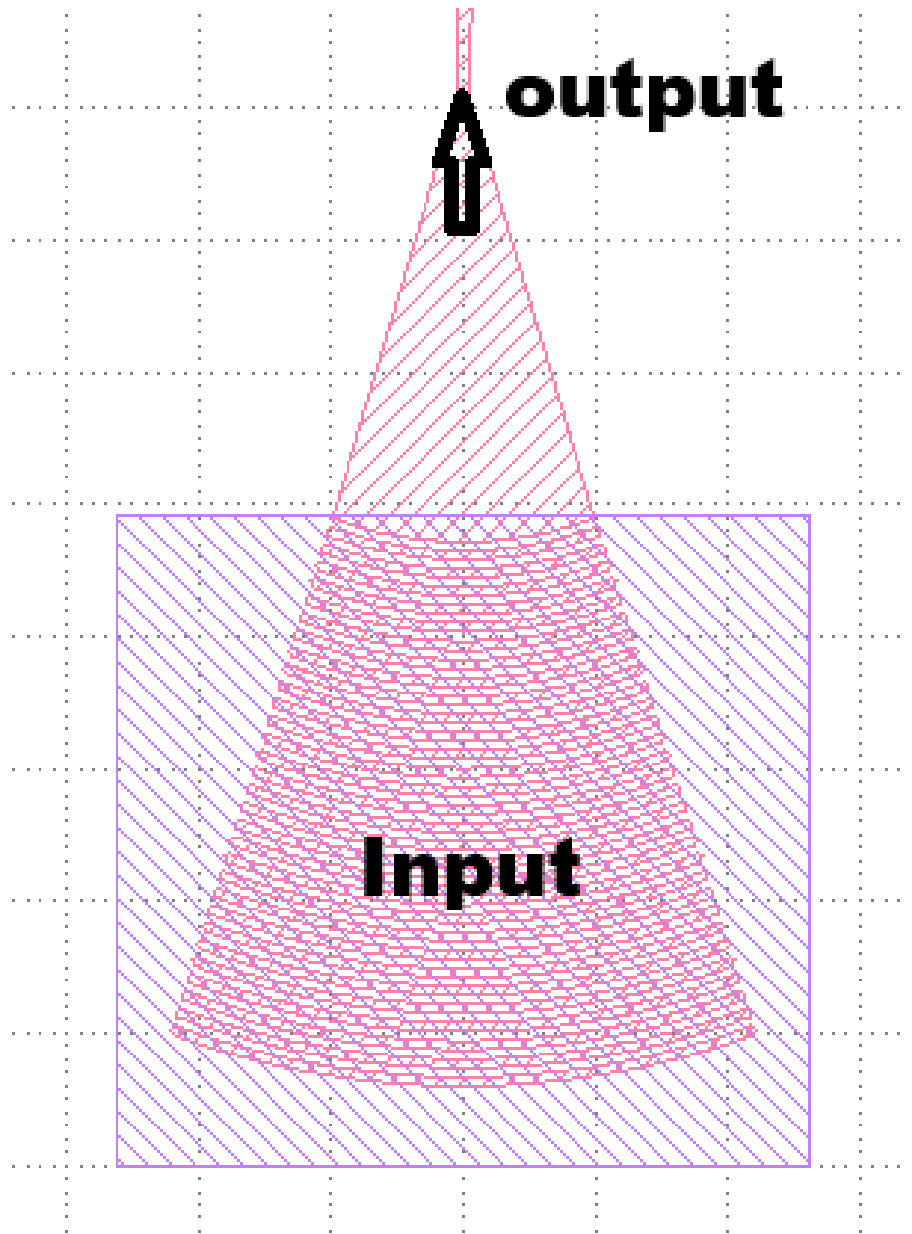


Figure 1: Grating Coupler tailored for $1550nm$ and 15° fiber angle, For Silicon and Silica platform.

2 Part 1: Mach Zender Interferometers

Mach Zender Interferometers (MZI) are photonic components that serve us as splitter and mergers of two different optical lengths with the purpose to set interference at the output due to phase difference by accumulation of a larger optical path at one of the arms, or changes on the effective indices of one of the paths. The output spectrum results with an interference pattern wavelength dependent which serves us for multiple purposes in optical sensing, digital and analog modulators for data communication, Spectrometers [1, 2], or as a simple phase shifter that serves a specific purpose on a chip.(See figure 2)

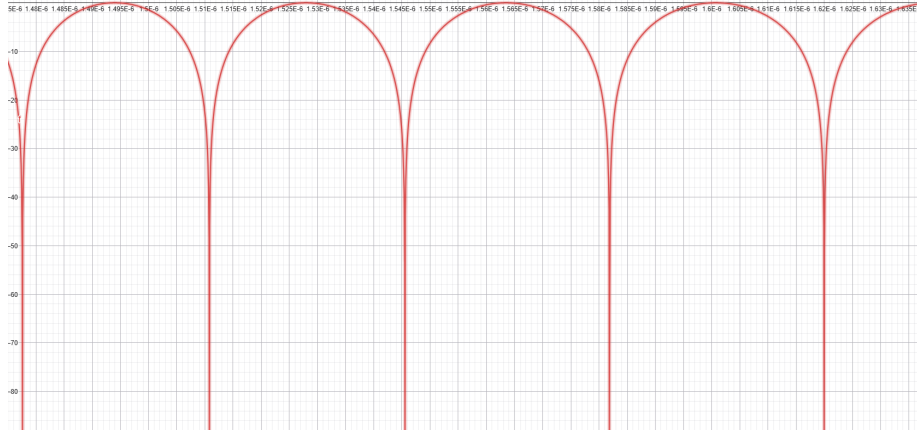


Figure 2: Output log spectrum example of an MZI with a path difference of $\Delta L \approx 27\mu m$, using Geogebra

This interferometers can be passive or active: active MZI requires external modulation like free carriers, thermal, or field modulation, we can, by using a proper PDK, build up a layers system that serves as an electrical modulator, where we can change systematically the effective index of the mode of one of the arms, and modulate the output intensity, and passive interferometers, which only uses difference path mechanism to create the fringes, or, by purpose, different fixed material environment as the usual with Silicon and Silica on one of the arms. The purpose of this part is to define mathematically and phisically the principle of functioning of passive type MZI.

For an unbalanced Mach-Zender Interferometer with spliting Y branch method at the input and output, we follow splitting, propagation and merging of the signals. We follow each step through the output to obtain the cosine expression of the spectrum fringes in log scale.

Considering lossless and optimized Y-splitters, the input phasor fiel would

be the sum of two fields with same amplitude, and, if Y-branch optimized [3, 4], in phase (see figure 3).

$$E_{in} = \frac{E_1}{\sqrt{2}} + \frac{E_2}{\sqrt{2}}$$

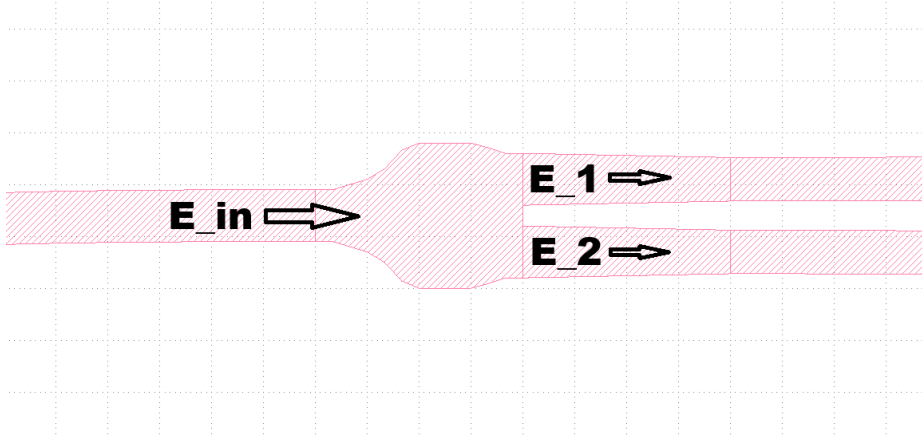


Figure 3: Y-branch Splitter showing fields in GDS

When E_1 and E_2 propagates along L_1 and L_2 (unbalanced), considering no propagation losses ($\alpha = 0$) then:

$$E_{1out} = \frac{E_1}{\sqrt{2}} e^{-i\beta L_1}$$

$$E_{2out} = \frac{E_2}{\sqrt{2}} e^{-i\beta L_2}$$

Upon arrival to the output, the fields recombine in a Y-branch merger, optimized to function as a merger taking into account the phase of each branch and minimizing losses. The total output would be:

$$E_{out} = E_{1out} + E_{2out}$$

$$E_{out} = \frac{E_1}{\sqrt{2}} e^{-i\beta L_1} + \frac{E_2}{\sqrt{2}} e^{-i\beta L_2}$$

Being $E_1 = E_2 = E_s$ thus:

$$E_{out} = \frac{E_s}{\sqrt{2}} (e^{-i\beta L_1} + e^{-i\beta L_2})$$

Let's find the intensity, by finding the squared norm of the phasor:

$$|E_{out}|^2 = \left| \frac{E_s}{\sqrt{2}} (e^{-i\beta L_1} + e^{-i\beta L_2}) \right|^2$$

At the entry $|E_s|^2 = \frac{I_{in}}{2}$

$$I_{out} = \frac{I_{in}}{4} (e^{-i\beta L_1} + e^{-i\beta L_2}) * (e^{i\beta L_1} + e^{i\beta L_2})$$

$$I_{out} = \frac{I_{in}}{2} (1 + \cos(\beta \Delta L))$$

This expression tells us the cosine behavior for MZIs output, on figure 4 is pictured the cosine function varying the path difference.

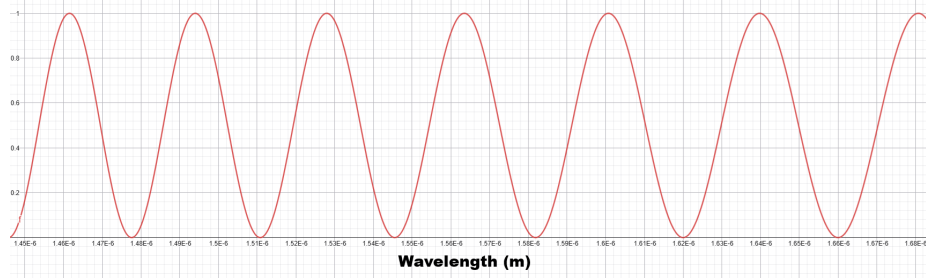


Figure 4: Output linear spectrum example of an MZI with a path difference of $\Delta L \approx 27\mu m$, using Geogebra

What is interesting is the variability of the transmittance VS wavelength, but the problem is the direct dependence of the n_{eff} with wavelength, there is a dispersion problem, so to calculate the bandwidth between continuous minimum transmittances (Free Spectral Range, FSR) is needed to take into account the dispersion, or in other words, the differential change of the transmittance phase $\Delta\phi$

$$\Delta\phi = \beta \Delta L = \frac{2\pi}{\lambda} n_{eff}(\lambda) \Delta L$$

Derivating the latter expression with respect to λ obtaining:

$$\frac{d\Delta\phi}{d\lambda} = \frac{d}{d\lambda} \left(\frac{2\pi}{\lambda} n_{eff}(\lambda) \Delta L \right)$$

$$\frac{d\Delta\phi}{d\lambda} = 2\pi \Delta L \frac{d}{d\lambda} \left(\frac{n_{eff}(\lambda)}{\lambda} \right)$$

$$\frac{d\Delta\phi}{d\lambda} = 2\pi \Delta L \left(\frac{1}{\lambda} \frac{dn_{eff}}{d\lambda} - \frac{n_{eff}}{\lambda^2} \right)$$

$$\frac{d\Delta\phi}{d\lambda} = -\frac{2\pi \Delta L}{\lambda^2} \left(n_{eff} - \lambda \frac{dn_{eff}}{d\lambda} \right)$$

We recognize $n_{group} = n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda}$:

$$\frac{d\Delta\phi}{d\lambda} = -\frac{2\pi\Delta L}{\lambda^2}n_g$$

To calculate the FSR we use the Taylor expansion of the transmittance phase, taking into account until the linear approximation of the phase and developing around the wavelength of use in the MZI.

$$\Delta\phi(\lambda) = \Delta\phi(\lambda_o) + \frac{d\Delta\phi(\lambda)}{d\lambda}|_{\lambda=\lambda_o}(\lambda - \lambda_o)$$

The phase difference between two consecutives zero transmittance wavelengths is -2π , hence:

$$\begin{aligned}\Delta\phi(\lambda_1) &= 0 - \frac{2\pi\Delta L}{\lambda_o^2}n_g(\lambda_1 - \lambda_o) \\ -2\pi &= -\frac{2\pi\Delta L}{\lambda_o^2}n_g(FSR)\end{aligned}$$

Finally:

$$FSR = \frac{\lambda_o^2}{n_g\Delta L}$$

Using a platform of $220nm$ thick and $420nm$ width we obtain, using mode solver, a group index of $n_g \approx 4.296053$ for $\lambda_o = 1550nm$ with the data, we find the ΔL needed for specific FSRs:

- $FSR = 20nm \rightarrow \Delta L \approx 27.962\mu m$
- $FSR = 50nm \rightarrow \Delta L \approx 11.185\mu m$
- $FSR = 100nm \rightarrow \Delta L \approx 5.592\mu m$

We will use these numbers to design our GDS.

3 Part 2: Ring Resonators

We will focus on the simple ring model to extract all the math formulas that governs the ring resonators behavior.

Ring resonators are incredible photonic estructures that are governed by the FP cavity principle [6, 7] where the trapping of light here is the physical principle we want to follow. Rings are cavities where we can fix specific number of wavelengths depending on the loss rates existing in coupling and propagation into the ring. See figure 5 to observe a typical one-bus ring-resonator also named as All-pass rings.

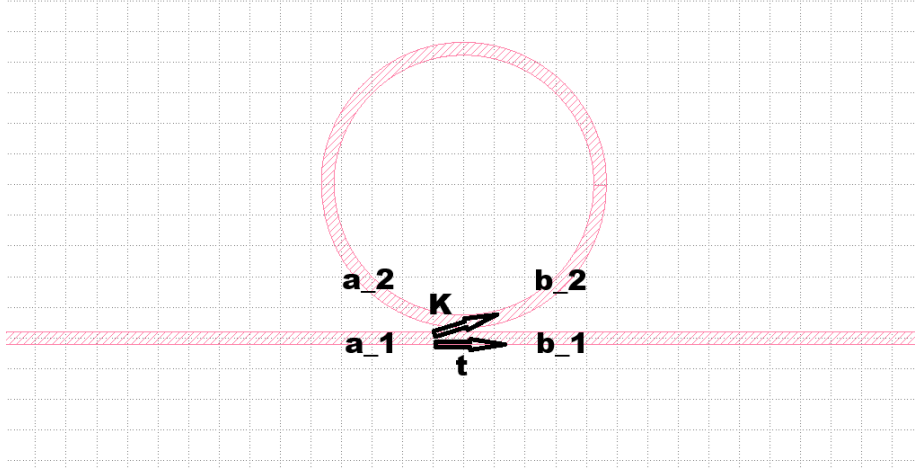


Figure 5: Ring resonator shown in a GDS

Being a_1, b_1, a_2, b_2 the coupling phasors of the porpagating light for each port (See figure 5). In the ring: $a_2 = b_2 e^{i\beta L - \frac{1}{2}\alpha L}$; and being $t, \kappa, t^*, -\kappa^*$ the self and cross coupling coefficients [7] seen the first two as the out-of-ring coefficients and the last two, out-of-bus coefficients. We can arrange all this as a 2x2 lienar system to solve for the output field phasor related with the input field phasor at the outter transmision port.

$$\begin{aligned} b_1 &= ta_1 + \kappa a_2 \\ b_2 &= -\kappa^* a_1 + t^* a_2 \end{aligned}$$

By managing the equations and looking an expression to relate the output phasor b_1 with the input phasor a_1 we yield the form to accurately know the phase delay between input and the output:

$$\frac{b_1}{a_1} = e^{i(\beta L + \pi)} \frac{A - t e^{i\beta L}}{1 - t^* A e^{i\beta L}}$$

Where $A = e^{-\frac{\alpha}{2}L}$ is the attenuation accumulated through the ring, those losses are related 100% to the ring conditions, its fabrication, its bending, and its material; t is related directly with the separation distance between the bus and the ring, also known as the gap, these separation will lead the self and cross coupling coefficients, these two numbers will totally set the behavior of the output spectrum, we can design by using the output spectrum desired.

The transmittance of the system could be calculated as the squared of the phasor relation:

$$\left|\frac{b_1}{a_1}\right|^2 = \frac{|t|^2 + A^2 - 2A|t|\cos(\beta L - \Phi)}{1 + |t|^2 A^2 - 2A|t|\cos(\beta L - \Phi)}$$

Being L the travelled distance in the ring, namely, perimeter of a circle. Where Φ is the phase shift (if applicable) of the self and cross coupling, could be considered as zero for no coupling phase shifting.

$$T = \frac{|t|^2 + A^2 - 2A|t|\cos(\beta L)}{1 + |t|^2 A^2 - 2A|t|\cos(\beta L)}$$

For zero transmittance we can obtain the phase condition where we can find the FSR for ring Resonators:

$$\cos(\beta L) = \frac{|t|^2 + A^2}{2A|t|}$$

If we would like 2π phase shift liakely as a MZI case, we need the right hand side of the latter equation equal 1:

$$\frac{|t|^2 + A^2}{2A|t|} = 1$$

$$|t|^2 - 2A|t| + A^2 = 0$$

By solving the last one finding the roots of the polynomial, we obtain the named Critical condition for Ring [7], to obtain a complete 2π shift between resonance wavelengths:

$$t = A$$

Namely, the self coupling coefficient must be equal to the propagation losses inside the ring, in other words, the amount of light coupled into the ring should be the same as the amount of light lost in it, this allows at the resonant wavelength perfectly destructive interference with the direct input light coupled onto the bus with the field leaked out the reing, total destruction of the fiel at the output for resonant wavelength, the wavelength is completely trapped into the ring.

Out of critical condition, there is still coupling but in a lesser sense, less trapping due to there is still light at resonant wavelength being able to pass through and interfering with the leaked field. This conditions are Under ($t > A$) and Over ($t < A$) coupling. At these conditions we obtain at linear domain fringes spectrum where the design conditions will determine the FSR, how depth and the quality of the resonance. See a representation of transmittance graph at critical coupling condition on figure 6.

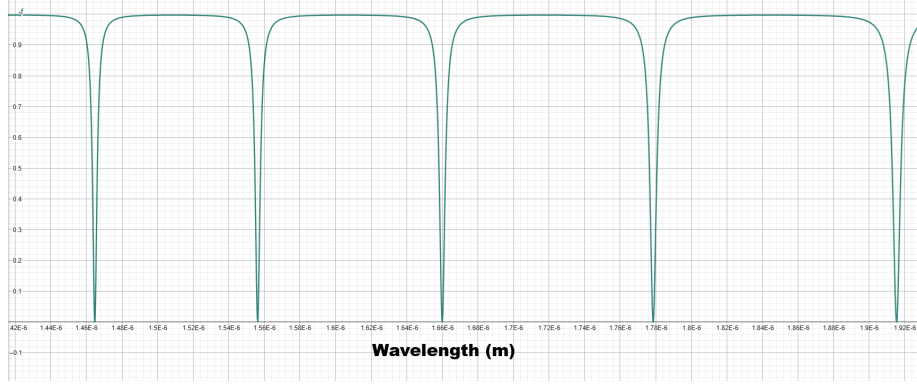


Figure 6: Linear ring output transmittance for $L \approx 10\mu m$, using Geogebra

For critical coupling, as we obtain the same phase difference condition as the MZI, the FSR would be the same:

$$FSR = \frac{\lambda^2}{2\pi R n_g}$$

Being R the radius of the ring resonator. By using this formula, we can design for specific Radius that generates FSRs desired. Using the same conditions of confinement, $220nm$ platform and $420nm$ width for single mode regime.

- $FSR = 20nm \rightarrow R \approx 4.45\mu m$
- $FSR = 50nm \rightarrow R \approx 1.78\mu m$
- $FSR = 100nm \rightarrow R \approx 0.89\mu m$

We have measured as well the gap por each example asuring critical coupling criteria. The procedure to accomplish this is the following is specified by Amsys based on Lumerical results. Normal rings technically have $0\mu m$ coupling length, but taking approach of the bent waveguide after gap position we can assure coupling using directional couplers criteria approximately [5].

$$t = \sqrt{\left(\frac{n_g \pi * L}{2Q\lambda}\right)^2 + 1} - \frac{n_g \pi * L}{2Q\lambda}$$

The last equation is the one obtained based on ring theory [5], where Q is the quality factor, calculated as the central wavelength of operation divided by the FWHM of the resonance dip. We will use the same FWHM used at [5] for simplicity, $0.8nm$. we can use it to estimate what would be the self-coupling coefficient to reach critical coupling criteria.

After that we use directional coupler theory to estimate what would be the coupling length, for our interest we find at [5] good results when the coupling length (L_c) is in the order of half the ring radius. So we will try to find those trying to follow fabrications DRCs and fixing wavelength, platform geometry and quality factor. Directional coupling theory demand the knowledge of difference between n_{even} and n_{odd} by fixing a gap distance. At the following are set the gap used for each FSR example assuring nearby critical coupling behaviour:

- $FSR = 20nm \rightarrow gap = 120nm$
- $FSR = 50nm \rightarrow gap = 90nm$
- $FSR = 100nm \rightarrow gap = 90nm$

For $\lambda = 1550nm$, $Q \approx 2000$, the same group index used in MZIs due to being the same platform, we use the following formula to calculate the coupling lengths:

$$L_c = \frac{\lambda}{\pi \Delta n} \sin^{-1}(\kappa)$$

To find κ we use the critical coupling criteria:

$$\begin{aligned} t &= A \\ \sqrt{1 - \kappa^2} &= t = A \\ \kappa &= \sqrt{1 - t^2} \end{aligned}$$

In the following table (see table 1) are summarized the results for each gap selection, using Modal analysis in Tidy3D we can find n_{even} and n_{odd} .

FSR (nm)	Radius (μm)	gap (nm)	Δn ($n_{even} - n_{odd}$)	L_c (μm)	Regime
20	4.45	120	0.08988	1.904	Critical
50	1.78	90	0.1174	0.933	Critical
100	0.89	90	0.1174	0.652	Over

Table 1: Summary of ring resonator design parameters for different FSR values.

Let's observe from table 1 that particularly for FSRs of $50nm$ and $100nm$, we designed for a gap of $90nm$ which for mayor foundries unrealizable, we can fix this by changing parameters allowing Δn to increase, this increasing the gap to fabricable widths, as well as varying the geometro for different platforms, or

varying the material, for example Silicon Nitride. Also the wavelength of use, we can tailor the grating for different wavelength of work, and for last but not least, modifying the Q factor as well. All these design parameters are the ones designers need to have into account to achieve real and useful photonic components.

4 Part 3: PDKs and DRCs

Process Design Kit (PDK)

A **Process Design Kit (PDK)** is a comprehensive collection of files, models, parameters, and design rules provided by a semiconductor foundry that enables designers to create photonic integrated circuits (PICs) compatible with a specific fabrication process [2].

The PDK serves as the critical interface between design and fabrication. It ensures that the structures designed in simulation and layout can be reliably manufactured with predictable performance. In silicon photonics, where device performance is highly sensitive to geometry and material properties, the PDK plays an essential role in bridging theoretical design and practical fabrication [2].

Role in Fabrication Compatibility

The PDK ensures fabrication compatibility in several ways:

- **Geometrical Constraints:** It defines minimum feature sizes, spacing, and alignment tolerances that are achievable by the fabrication process.
- **Process Variability Awareness:** It includes models that account for fabrication variations such as etch depth deviations or sidewall roughness.
- **Technology Mapping:** It maps abstract design elements (e.g., waveguides, couplers) to actual mask layers used in fabrication.

By enforcing these constraints, the PDK reduces the risk of fabrication failure and ensures that the fabricated device behaves close to its simulated performance [2].

Standardized Layers and Design Rules

A key function of the PDK is the definition of **standardized layers** and **design rules**. These include:

- Layer definitions corresponding to physical fabrication steps.
- Rules governing how layers interact (e.g., overlaps, spacing).
- Parameterized cells (PCells) for reusable photonic components.

This standardization enables scalable and modular PIC design, similar to electronic IC design.

Generic PDK in GDSFactory

The generic PDK provided in GDSFactory includes a set of predefined layers that represent different fabrication steps [8]. The main categories are described below.

Active/Core Layers These layers define regions where optical confinement occurs, typically corresponding to high-index materials such as silicon.

- Used to define waveguide cores.
- Strongly influence effective index and optical mode confinement.
- Critical for determining propagation characteristics.

Passive Waveguide Layers These layers represent waveguides that do not include active modulation or gain.

- Used for routing optical signals.
- Often correspond to partially etched or fully etched silicon regions.
- Designed to minimize propagation loss.

Metal Interconnect Layers These layers are used for electrical routing and contact formation.

- Enable integration of photonic and electronic components.
- Used in modulators, heaters, and photodetectors.
- Typically include multiple metal levels for routing flexibility.

Etch Layers Etch layers define regions where material is removed during fabrication.

- Full etch: removes material completely down to the substrate.
- Partial etch: creates rib or ridge waveguides.
- Control optical confinement and scattering losses.

Implant Layers These layers specify regions where dopants are introduced.

- Used to create p-type and n-type regions.
- Essential for active devices such as modulators and photodetectors.
- Affect carrier concentration and optical absorption.

Other Layers Additional layers may include:

- **Heater layers:** for thermo-optic tuning.
- **Via layers:** for electrical connections between metal levels.
- **Cladding/opening layers:** for defining oxide regions or openings.

Overall, the generic PDK in GDSFactory provides a simplified but representative abstraction of a real fabrication process, allowing designers to develop layouts consistent with foundry constraints [8].

Design Rule Check (DRC)

A **Design Rule Check (DRC)** is a verification process that ensures a layout complies with the geometric and topological constraints defined in the PDK [2].

DRC verifies that all features in the design satisfy fabrication limits, such as minimum widths, spacings, and layer overlaps. It is a crucial step before tape-out, as violations may lead to fabrication errors or degraded device performance.

Importance in PIC Design

In photonic integrated circuits, DRC is particularly important due to the sensitivity of optical behavior to geometry. Small deviations can lead to:

- Increased propagation losses,
- Shifted resonance wavelengths,
- Reduced coupling efficiency,
- Device failure.

Thus, DRC ensures both **fabricability** and **functional reliability** [2].

DRC in GDSFactory

In GDSFactory, DRC is applied using rules defined in the PDK. These rules are checked automatically against the layout [8].

Typical checks include:

- Minimum feature size verification,
- Minimum spacing between structures,
- Layer overlap and enclosure rules,
- Alignment between different process layers.

The tool flags violations, allowing the designer to correct them before fabrication.

Examples of Typical DRC Violations

Minimum Width Violation

- Occurs when a waveguide width is smaller than the allowed minimum.
- **Impact:** Increased scattering loss or complete fabrication failure.

Minimum Spacing Violation

- Occurs when two structures are placed too close together.
- **Impact:** Unintended optical coupling or fabrication merging.

Enclosure Violation

- Occurs when one layer does not sufficiently cover another (e.g., metal over contact).
- **Impact:** Poor electrical connectivity or reliability issues.

Overlap Violation

- Occurs when layers overlap incorrectly.
- **Impact:** Fabrication ambiguity or incorrect device formation.

Impact on Device Performance

DRC violations can significantly degrade device performance:

- **Optical Loss:** Caused by rough or improperly defined waveguides.
- **Resonance Shift:** Due to dimensional inaccuracies in resonators.
- **Coupling Errors:** Resulting from incorrect spacing in couplers.
- **Electrical Failure:** Due to improper metal connections.

Ensuring a clean DRC is therefore essential not only for fabrication success but also for achieving the desired optical functionality.

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